

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-1

Course: ECE/EEE F311 Communication Systems
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Objectives

- Understand the concept of block-diagram approach, transmission bandwidth, and Low pass filtering to input signals.

Notations

$$\text{sinc}(x) = \frac{\sin(x)}{x}.$$

Tutorial Problems

1. A signal $m(t) = 8000\text{sinc}(8000\pi t)$ is passed through a channel with impulse response (i) $h(t) = 9000\text{sinc}(9000\pi t)$ (ii) $h(t) = 4000\text{sinc}(4000\pi t)$. Discuss qualitatively about the output signal for both the cases.
2. The human voice signal typically contains significant frequency components up to approximately 3.3 kHz. To model this behavior mathematically, consider a signal $m(t)$ composed of $N = 30$ harmonically related sinusoidal components with a fundamental frequency of $f_0 = 110$ Hz. The signal is defined as

$$m(t) = \sum_{k=1}^{30} \frac{1}{k} \cos(2\pi k f_0 t).$$

- (a) Compute the average power of the signal $m(t)$.
- (b) Determine a finite-duration speech segment by applying a rectangular window of width $t_0 = 10$ ms to produce the truncated signal

$$m_w(t) = m(t)w(t), \quad \text{where} \quad w(t) = \begin{cases} 1, & 0 \leq t \leq t_0, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, determine the effective bandwidth of the windowed signal $m_w(t)$ by estimating the main lobe width bandwidth of the rectangular window.

- (c) A channel $h(t) = 8000\text{sinc}(8000\pi t)$ is available for transmission. Draw the transmitter block diagram.

3. In digital communication systems, information is transmitted as a stream of binary digits such as 01100011. . . . For physical transmission over a channel, each bit—'1' or '0'—must be represented by a corresponding waveform. Let bit '1' be represented by the signal $m_1(t)$ and bit '0' by $m_0(t)$. Model the digital transmission waveform by appropriately mapping the bit stream into a continuous-time signal. Assume $m_1(t)$ as a rectangular pulse of width T_1 with amplitude $+A$ and $x_0(t)$ as rectangular pulse of width T_0 with amplitude $-A$. Compute the average energy per bit E_b . Repeat the same analysis for the case where $m_1(t) = A\text{sinc}(\pi t/T_1)$ and $m_0(t) = -A\text{sinc}(\pi t/T_0)$.
4. You are transmitting a voice signal $m(t)$ to a friend in Delhi using a landline telephone system. The two telephone systems are interconnected via a twisted pair cable. Assume that the overall communication system can be modeled as a linear time-invariant (LTI) system, where the transmitter has an impulse response denoted by $g_1(t)^{\text{tx}}$, the transmission channel by $h(t)$, and the receiver by $g_2(t)^{\text{rx}}$. Mathematically express the received signal $\hat{m}(t)$ as a convolution of these components. Design an appropriate receiver filter to optimally recover the original message signal $m(t)$, and discuss the practical challenges that may arise during signal detection.

Solution

1. Using the Fourier transform pair

$$2B \operatorname{sinc}(2\pi Bt) \longleftrightarrow \Pi\left(\frac{f}{2B}\right),$$

the signal $m(t) = 8000 \operatorname{sinc}(8000\pi t)$ has a bandwidth of 4000 Hz.

- **(i):** The channel has a bandwidth of 4500 Hz, the signal's entire frequency content is preserved. The signal passes through the channel without any frequency distortion.
- **(ii):** The channel bandwidth is limited to 2000 Hz, the higher frequency components of the signal (above 2000 Hz) are attenuated. This truncation of the spectrum causes distortion in the received signal due to the loss of high-frequency content.

2. (a) To compute the average power of a continuous-time signal, we use the formula

$$P_{\text{avg}} = \frac{1}{T_0} \int_0^{T_0} x^2(t) dt.$$

For a sinusoidal signal of the form $A \cos(2\pi f_c t)$, the average power is given by $A^2/2$. In the case of the signal

$$m(t) = \sum_{k=1}^{30} \frac{1}{k} \cos(2\pi k f_0 t),$$

each harmonic component has an amplitude $1/k$, and thus contributes an average power of $1/(2k^2)$. The total average power is the sum of the powers of all harmonics:

$$P_{\text{avg}} = \sum_{k=1}^{30} \frac{1}{2k^2} = \frac{1}{2} \sum_{k=1}^{30} \frac{1}{k^2}.$$

- (b) The FT pair of the symmetric rectangular pulse:

$$\Pi\left(\frac{t}{T}\right) \leftrightarrow T \operatorname{sinc}(\pi f T)$$

The rectangular window function:

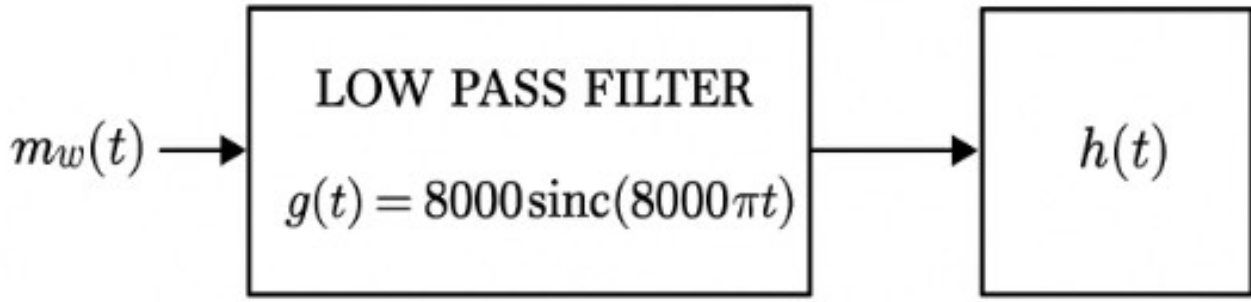
$$w(t) = \Pi\left(\frac{t - \frac{t_0}{2}}{t_0}\right)$$

FT:

$$\mathcal{F}\left\{\Pi\left(\frac{t - \frac{t_0}{2}}{t_0}\right)\right\} = t_0 \cdot \operatorname{sinc}(\pi f t_0) \cdot e^{-j\pi f t_0}$$

Thus the bandwidth of the main lobe of $W(f) = 1/t_0 = 100\text{Hz}$. The highest frequency content of $m(t)$: 3300 Hz. Multiplication in time-domain leads to convolution in the frequency domain. Thus, the BW of $m_w(t)$ is $3300 + 100 = 3400$ Hz.

- (c) The message signal $m_w(t)$, which has infinite bandwidth, is passed through a low-pass filter $g(t) = 8000 \operatorname{sinc}(8000\pi t)$ in order to limit its bandwidth and make it compatible with the channel $h(t)$.



3. Denote T_b as the bit period.

$$x(t) = \sum_{k=0}^{\infty} m_{b_k}(t - kT_b), \quad \text{where} \quad m_{b_k}(t) = \begin{cases} m_1(t), & b_k = 1 \\ m_0(t), & b_k = 0 \end{cases}$$

$$E_1 = \int_{-\infty}^{\infty} |m_1(t)|^2 dt = \int_0^{T_1} A^2 dt = A^2 T_1$$

$$E_0 = \int_{-\infty}^{\infty} |m_0(t)|^2 dt = \int_0^{T_0} A^2 dt = A^2 T_0$$

Assuming equal probability of bits '0' and '1', the average energy per bit is:

$$E_b = \frac{1}{2}E_1 + \frac{1}{2}E_0 = \frac{1}{2}A^2T_1 + \frac{1}{2}A^2T_0 = \frac{A^2}{2}(T_1 + T_0)$$

If $T_1 = T_0$, then $E_b = A^2$.

For the sinc function (It can be left for take home if time is limited):

$$m(t) = A \cdot \text{sinc}\left(\frac{\pi t}{T_1}\right)$$

$$M(f) = AT_1 \cdot \text{rect}(fT_1)$$

$$E_1 = \int_{-\infty}^{\infty} |M(f)|^2 df = A^2 T_1^2 \int_{-1/(2T_1)}^{1/(2T_1)} 1 df = A^2 T_1$$

Similarly

$$E_0 = \int_{-\infty}^{\infty} |M(f)|^2 df = A^2 T_0^2 \int_{-1/(2T_0)}^{1/(2T_0)} 1 df = A^2 T_0$$

Assuming equal probability of bits '0' and '1', the average energy per bit is:

$$E_b = \frac{1}{2}E_1 + \frac{1}{2}E_0 = \frac{1}{2}A^2T_1 + \frac{1}{2}A^2T_0 = \frac{A^2}{2}(T_1 + T_0)$$

If $T_1 = T_0$, then $E_b = A^2$.

4. Assume the voice signal $m(t)$ passes through a linear time-invariant (LTI) system comprising a transmitter $g_1(t)^{\text{tx}}$, a channel $h(t)$, and a receiver filter $g_2(t)^{\text{rx}}$. The received signal is given by the convolution:

$$\hat{m}(t) = m(t) * g_1(t)^{\text{tx}} * h(t) * g_2(t)^{\text{rx}} = m(t) * g_{\text{eq}}(t),$$

where $g_{\text{eq}}(t) = g_1(t)^{\text{tx}} * h(t) * g_2(t)^{\text{rx}}$ is the overall system impulse response.

Taking the Fourier Transform of all components, we get:

$$\hat{M}(f) = M(f) \cdot G_1(f)^{\text{tx}} \cdot H(f) \cdot G_2(f)^{\text{rx}}.$$

To recover the original message signal $m(t)$, we require $\hat{M}(f) = M(f)$, which implies:

$$G_1(f)^{\text{tx}} \cdot H(f) \cdot G_2(f)^{\text{rx}} = 1.$$

Thus, the ideal receiver filter should be designed as:

$$G_2(f)^{\text{rx}} = \frac{1}{G_1(f)^{\text{tx}} \cdot H(f)}.$$

In practice, several challenges arise:

- The channel response $H(f)$ and transmit filter $G_1(f)^{\text{tx}}$ are unknown.
- If $|H(f)|$ is small at some frequencies, inversion will be a problem.